Module IV

Market Structure

Unit 18: Perfect Competition

I. Price Taking

II. Why the Firm's Demand Curve Is Horizontal

III. Derivation of a Competitive Firm's Demand Curve

IV. Profit Maximization

V. Short-Run Competitive Profit Maximization

VI. Long-Run Competitive Profit Maximization

Objectives of the Study:

After studying this unit, the student should be able to

- understand the market structure Perfect Competition

Perfect Competition

Perfect competition is a common market structure that has very desirable properties, so it is useful to compare other market structures to competition. In this section, we describe the properties of competitive firms and markets.

In general terms, competition refers to firms rivalling with one another to attract customers. However, in economics, a market is considered perfectly competitive only when firms are price takers, meaning they cannot influence market prices and must accept the price determined by market demand and supply.

Perfect competition represents an ideal market situation where resources are allocated efficiently, and no individual buyer or seller has control over the price.

I. Price Taking

When most people talk about "competitive firms," they usually refer to firms that compete or rival with one another to attract customers. In this general sense, any market with more than

one firm may be considered competitive. However, in economics, the term “competitive market” has a more precise meaning.

Economists define a market as competitive only when all firms within it are price takers. A price-taking firm is one that cannot influence the market price of its product or the prices of the inputs it uses. Instead, the price is determined by the overall interaction of market demand and supply, and each individual firm must accept this price.

Why is a firm a price taker?

A firm becomes a price taker because of the structure of the market in which it operates. In a perfectly competitive market:

- There are a large number of firms.
- Each firm produces only a very small portion of total output.
- Products are identical across firms.

Due to these conditions, no single firm has enough market power to influence price. If one firm changes its output, the effect on total market supply is negligible, and therefore the market price remains unchanged.

Role of the Demand Curve

A key reason why firms are price takers is that they face a perfectly elastic (horizontal) demand curve at the prevailing market price. This has several important implications:

- The firm can sell any quantity of output at the given market price.
- The firm has no incentive to lower its price, because it can already sell all its output at the existing price.
- The firm cannot increase its price, because buyers would immediately switch to other sellers offering the same product.

Infinitely Elastic Demand

The demand curve faced by the firm is said to be infinitely elastic. This means:

- Even a very small increase in price leads to a complete loss of customers.

- At the market price, demand is perfectly responsive, allowing the firm to sell any quantity.

For example, consider a farmer selling wheat in a large agricultural market. If the farmer tries to charge a higher price than others, buyers will simply purchase wheat from other farmers since all wheat is identical. As a result, the farmer has no control over price and must accept the prevailing market rate.

II. Why the Firm's Demand Curve Is Horizontal

In a perfectly competitive market, the demand curve faced by an individual firm is perfectly elastic, meaning it is a horizontal straight line at the prevailing market price. This occurs because firms are price takers and cannot influence the price through their individual actions.

The horizontal nature of the demand curve is not accidental; it is the result of several important characteristics of perfect competition.

1. Large Number of Buyers and Sellers

In a perfectly competitive market, there are a large number of buyers and sellers, each of whom is very small relative to the overall market. As a result:

- No single firm can influence the total market supply.
- Any change in the output of a single firm has a negligible effect on market price.

For example, soybean farmers are price takers because each farmer produces only a tiny fraction of total output. Even if one farmer stops producing, the overall market supply changes very little, so the market price remains unaffected.

Thus, each firm must accept the given market price, which leads to a horizontal demand curve.

2. Identical (Homogeneous) Products

All firms in perfect competition produce identical or homogeneous products. Consumers perceive no difference between goods supplied by different firms.

- A buyer does not care which firm supplies the product.

- If one firm charges a higher price, consumers will immediately switch to other sellers.

For instance, agricultural products like apples or wheat are considered identical in quality. Because of this uniformity, no firm can charge a price above the market level.

This forces firms to accept the market price, making the demand curve perfectly elastic.

3. Full Information

In a perfectly competitive market, buyers and sellers have complete knowledge about:

- Prices charged by all firms
- Product quality and characteristics

This transparency ensures that:

- Buyers always purchase from the cheapest seller.
- Firms cannot charge higher prices without losing all customers.

If a firm attempts to increase its price, consumers—being fully informed—will shift instantly to other firms. Hence, the firm's demand drops to zero above the market price.

4. Negligible Transaction Costs

Transaction costs refer to the costs involved in buying or selling a product, such as time, transportation, or legal expenses.

In perfect competition:

- These costs are very low or negligible.
- Buyers can easily switch from one seller to another without any inconvenience.

For example, in commodity markets, buyers can quickly compare prices and make purchases with minimal effort. Because switching is so easy, even a slight increase in price by a firm will result in a complete loss of demand. This reinforces the horizontal shape of the demand curve.

5. Free Entry and Exit of Firms

Firms can freely enter or exit the market depending on profit conditions:

- If firms earn abnormal profits, new firms enter the market.
- Increased competition pushes prices down to the normal level.
- If firms incur losses, some firms exit, reducing supply and stabilizing prices.

This process ensures that:

- No firm can sustain a higher price in the long run.
- Firms remain price takers at all times.

A practical example of a nearly perfectly competitive market is the Chicago Mercantile Exchange.

This market exhibits the following features:

- Thousands of buyers and sellers participate in trading.
- Products such as wheat and other commodities are standardized and identical.
- Prices are publicly available, ensuring full information.
- Transactions occur quickly with minimal cost.
- Entry and exit into trading are relatively easy.

Because of these conditions, traders in such markets cannot influence prices individually and must accept the prevailing market price.

Deviations from Perfect Competition

In reality, very few markets satisfy all the strict conditions of perfect competition. Most markets deviate from this ideal structure in one way or another. However, even when these deviations exist, many markets still behave in a manner similar to perfectly competitive markets, especially when there are a large number of firms and strong competition.

1. Government Restrictions on Entry

In some markets, the government imposes restrictions that limit the entry of new firms. These restrictions may include:

- Licensing requirements
- Zoning laws
- Legal regulations

Such barriers prevent completely free entry and exit, which is a key feature of perfect competition. However, if the number of existing firms is still large, individual firms may continue to behave as price takers because no single firm has significant control over price.

2. Presence of Transaction Costs

In the real world, transaction costs are rarely zero. Buyers and sellers may incur costs such as:

- Transportation expenses
- Search costs (finding the best price)
- Legal or administrative costs

When transaction costs are high, consumers may not switch sellers even if there is a price difference. This gives some firms limited pricing power. However, if transaction costs are relatively low, markets can still function close to perfect competition.

3. Incomplete or Imperfect Information

Perfect competition assumes that all buyers and sellers have complete information about prices and products. In reality:

- Consumers may not know all available prices
- Information about product quality may be limited

Despite this, even if only some buyers are well-informed, they can influence the market by choosing lower-priced sellers. This competitive pressure prevents firms from significantly increasing prices.

4. Product Differentiation

In many markets, products are not perfectly identical. Firms may differentiate their products through:

- Branding
- Quality differences
- Packaging

This reduces the degree of competition and gives firms some control over pricing. However, if products are still close substitutes, competition remains strong.

5. Limited Mobility of Resources

Factors of production such as labor and capital may not move freely due to:

- Skill differences
- Geographical constraints
- Institutional barriers

This restricts the flexibility assumed in perfect competition. Even so, in industries where mobility is relatively high, firms behave more like price takers.

III. Derivation of a Competitive Firm's Demand Curve

The demand curve faced by an individual firm in a competitive market is derived from the concept of residual demand. Residual demand refers to the portion of total market demand that remains after accounting for the supply provided by all other firms in the market.

In other words, a firm can only sell what other firms do not supply at a given price.

Residual Demand Concept

A firm faces a residual demand curve defined as:

$$D_r(p) = D(p) - S_o(p)$$

Where:

- $D_r(p)$ = Residual demand for the firm
- $D(p)$ = Total market demand

- $S_o(p)$ = Supply of all other firms

This equation shows that the demand available to an individual firm depends on both the total market demand and the quantity supplied by competing firms.

The concept of residual demand can be understood step by step:

- A firm sells its product only to those consumers who have not already purchased from other firms.
- At high prices, other firms are willing to supply a large quantity. If their supply fully satisfies market demand, the residual demand for the firm becomes zero.
- At lower prices, other firms supply less than the total market demand. The remaining unmet demand becomes available to the individual firm.

Thus, the firm's demand is not the total market demand, but only the leftover demand after other firms have supplied their share.

The residual demand curve is derived by comparing:

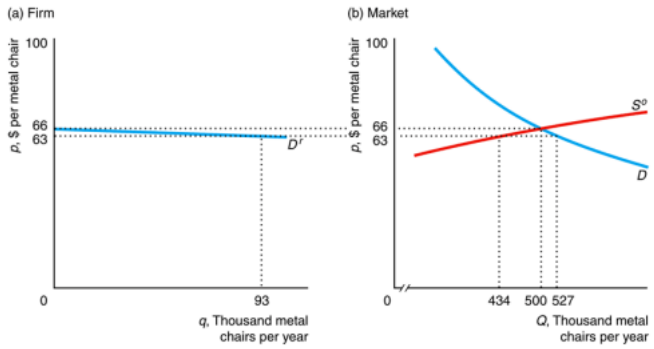
- The market demand curve, and
- The supply curve of other firms

At each price level:

- Subtract the quantity supplied by other firms from total market demand.
- The remaining quantity represents the demand faced by the individual firm.

Graphically, the residual demand curve is the horizontal difference between the market demand curve and the supply curve of other firms as shown in the figure 18.1.

Figure 18.1: Residual Demand Curve



The market demand that is not met by other sellers at any given price. The firm's residual demand function, Dr(p), shows the quantity demanded from the firm at price p. A firm sells only to people who have not already purchased the good from another seller. We can determine how much demand is left for a particular firm at each possible price using the market demand curve and the supply curve for all other firms in the market. The quantity the market demands is a function of the price:

$$Q = D(p)$$

The supply curve of the other firms is $S_o(p)$. The residual demand function equals the market demand function, D(p), minus the supply function of all other firms:

$$D_r(p) = D(p) - S_o(p)$$

At prices so high that the amount supplied by other firms, $S_o(p)$, is greater than the quantity demanded by the market, *D*(*p*), the residual quantity demanded, $D_r(p)$, is zero. In Figure 18.1 we derive the residual demand for a Canadian manufacturing firm that produces metal chairs. Panel b shows the market demand curve, D, and the supply of all but one manufacturing firm, S_o. At p = $66 per chair, the supply of other firms, 500 units (where one unit is 1,000 metal chairs) per year, exactly equals the market demand (panel b), so the residual quantity demanded of the remaining firm (panel a) is zero. At prices below $66, the other chair manufacturers are not willing to supply as much as the market demands. At p = $63, for example, the market demand is 527 units, but other firms want to supply only 434 units. As a result, the residual quantity demanded from the individual firm at p = $63 is 93 (= 527 - 434) units. Thus, the residual demand curve at any given price is the horizontal difference between the market demand curve and the supply curve of the other firms. The residual demand curve the firm faces in panel a is much flatter than the market demand curve in panel b. As a result, the elasticity of the residual demand curve is much higher than the market elasticity. If the market has n identical firms, the elasticity of demand, ε_i, facing Firm i is

$$\varepsilon_i = n\varepsilon - (n - 1)\eta_o,$$

where ε is the market elasticity of demand (a negative number), η_o is the elasticity of supply of each of the other firms (typically a positive number), and n - 1 is the number of other firms. As equation $\varepsilon_i = n\varepsilon - (n - 1)\eta_o$ shows, a firm's residual demand curve is more elastic the more firms, n, are in the market, the more elastic the market demand, ε, and the larger the elasticity of supply of the other firms, η_o. The residual demand elasticity, ε_i, must be at least as elastic as $n\varepsilon$ if the supply curve slopes up so that the second term makes the estimate more elastic. Thus, using $n\varepsilon$ as an approximation is conservative. For example, even though the estimated market elasticity of demand for soybeans is very inelastic at about $\varepsilon = -0.2$, because n = 107,000, the residual demand facing a single soybean farm must be at least $n\varepsilon = 107{,}000 \times (-0.2) = -21{,}400$, which is extremely elastic.

Thus, we can say that the residual demand curve has the following characteristics:

- It is flatter (more elastic) than the market demand curve.
- As the number of firms increases, each firm's share of the market becomes smaller.
- Consequently, the residual demand curve becomes more elastic.
- In the case of a very large number of firms, the curve becomes perfectly horizontal, indicating perfectly elastic demand.

This explains why firms in perfect competition are price takers.

IV. Profit Maximization

Economists generally assume that all firms aim to maximize profit. This assumption is based on two main reasons:

- Many businesspeople explicitly state that profit maximization is their objective.
- Firms that fail to maximize profit are likely to incur losses and eventually be forced out of the market.

Thus, profit maximization is considered the primary goal of firms in economic theory.

Meaning of Profit

A firm's profit π, is defined as the difference between its total revenue R and total cost C:

$$\pi = R - C$$

- If ($\pi > 0$) → Profit
- If ($\pi < 0$) → Loss

Economic Cost vs Business Cost

Economists measure cost as opportunity cost, which includes:

- Explicit costs (wages, materials, rent)
- Implicit costs (forgone income, owner's time)

This leads to a difference between:

- Business profit (accounting profit)
- Economic profit

Example:

Suppose a person earns ₹20,000 from their own business. If they could earn ₹25,000 working elsewhere, then:

- Economic profit = 20,000 − 25,000 = –₹5,000 (loss)

Thus, ignoring opportunity cost can lead to incorrect decisions.

A firm must make two key decisions for profit maximisation:

1. Output Decision

What level of output q* maximizes profit?

2. Shutdown Decision

Should the firm produce or shut down?

Profit Maximization Rules

Rule 1: The firm produces the output level where profit is maximum.

Rule 2: The firm produces where marginal profit = 0.

Rule 3: MR(q) = MC(q)

Where:

- MR = Marginal Revenue
- MC = Marginal Cost

That is,

- If MR > MC → Increase output
- If MR < MC → Reduce output
- If MR = MC → Profit is maximized

Shutdown Rule

The shutdown rule is an important concept in economics that helps a firm decide whether it should continue production or temporarily cease operations in order to minimize losses. This decision depends on the relationship between the firm's revenue and its costs, and it differs between the short run and the long run.

In the short run, a firm faces two types of costs namely fixed costs and variable costs. Fixed costs, such as rent, machinery, and equipment, are considered sunk costs because they must be paid regardless of whether the firm produces any output. Since these costs cannot be recovered, they are irrelevant when making the shutdown decision. Instead, the firm focuses only on variable costs, which include expenses like wages, raw materials, and electricity—costs that can be avoided if production stops. If the firm's total revenue is greater than or equal to its variable cost, it should continue producing, even if it is making an overall loss. This is because the firm is covering all its variable costs and contributing something toward its fixed costs, thereby reducing its total loss. However, if the firm's revenue falls below its variable cost, it means that the firm cannot even cover the costs that could be avoided by shutting down. In such a situation, continuing production would increase losses, so the firm should temporarily shut down. So in the short run shut down only if: Revenue < Variable Cost.

In the long run, the shutdown decision becomes simpler because all costs are variable and avoidable. There are no fixed or sunk costs, as the firm has enough time to adjust all inputs, sell assets, or exit the industry completely. Therefore, the firm must ensure that its total revenue

covers all its costs. If the firm is unable to do so and incurs any economic loss, it will choose to shut down and exit the market. Unlike in the short run, there is no justification for continuing production at a loss because all costs can be avoided.

V. Short-Run Competitive Profit Maximization

Meaning of Short Run

The short run in economics refers to a period of time during which at least one factor of production remains fixed and cannot be changed. This fixed factor is usually capital, such as machinery, buildings, or equipment, which cannot be increased or reduced quickly due to time and cost constraints. Because of this limitation, firms can adjust only some inputs—like labor or raw materials—while others remain constant. In the short run, firms also cannot freely enter or exit the market. New firms cannot quickly enter because setting up production facilities takes time, and existing firms cannot completely leave the market since they are tied to fixed assets. However, they may choose to temporarily stop production if conditions are unfavourable. Thus, the short run is characterized by limited flexibility in production decisions and constraints on market participation.

Output Decision in Short Run

In the short run, a firm operating under perfect competition determines its level of output based on the goal of profit maximization. Since the firm is a price taker, it has no control over the market price and must accept the price determined by market demand and supply. As a result, the firm's price (P) is equal to its marginal revenue (MR). This means that every additional unit sold adds the same amount to revenue, equal to the market price. Therefore, the profit-maximizing condition for a competitive firm becomes:

$$MC(q) = p$$

This condition implies that the firm will adjust its level of output until its marginal cost (MC) equals the market price. If the marginal cost of producing an additional unit is less than the price (MC < P), the firm can increase its profit by producing more, because the revenue from selling that unit exceeds its cost. On the other hand, if the marginal cost is greater than the price (MC > P), the firm should reduce its output, as producing extra units would add more to cost than to revenue, thereby reducing profit. Profit is maximized only when MC = P, because at

this point, the additional revenue from producing one more unit exactly equals the additional cost, and there is no further gain from changing output. Thus, in the short run, a competitive firm carefully adjusts its production level to this equilibrium condition in order to achieve maximum profit or minimum loss.

Profit or Loss in the Short Run and Short-Run Supply Curve

In the short run, a firm operating under perfect competition may find itself in different profit situations depending on market conditions and its cost structure. The firm may earn supernormal (abnormal) profit when its total revenue exceeds total cost, or normal profit when revenue just equals total cost, covering both explicit and implicit costs. However, the firm may also incur losses when total cost exceeds total revenue. Despite this possibility, a firm does not immediately shut down when it incurs losses in the short run. This is because fixed costs, such as rent and machinery, are unavoidable in the short run. As long as the firm's revenue is sufficient to cover its variable costs, it will continue production, since doing so allows it to cover part of its fixed costs and thereby reduce overall losses. The firm will only shut down if its revenue falls below variable costs.

The short-run supply curve of a competitive firm is derived from its marginal cost (MC) curve, but only the portion that lies above the average variable cost (AVC) curve. This is because the firm will produce only when price (which equals marginal revenue) is at least equal to AVC. If the price falls below AVC, the firm cannot cover its variable costs and will shut down. Thus, the MC curve above AVC represents the firm's willingness to supply output at different price levels.

The key insight from this analysis is that in the short run, firms have limited flexibility due to fixed costs. As a result, they may continue operating even when they are making losses, provided they can cover their variable costs. Fixed costs cannot be avoided in the short run, so firms focus on minimizing losses rather than immediately exiting the market.

VI. Long-Run Competitive Profit Maximization

Meaning of Long Run

The long run in economics refers to a period of time sufficiently long for a firm to adjust all its factors of production. Unlike the short run, where at least one factor remains fixed, in the long

run all inputs are variable, meaning firms can change the scale of production by increasing or decreasing labor, capital, machinery, and plant size. Because of this flexibility, firms can fully adapt to changes in market conditions.

Another important feature of the long run is that firms can freely enter or exit the market. If existing firms are earning profits, new firms are attracted to the industry, increasing competition. On the other hand, if firms are incurring losses, some will leave the market. This process continues until only normal profits are earned.

In the long run, no fixed costs remain, because all costs can be adjusted or avoided. Firms are not bound by past investments, and they can reorganize production or completely shut down if it is not profitable. As a result, the long run represents a period of full flexibility and adjustment, where firms make decisions that ensure efficiency and long-term sustainability.

Output Decision in Long Run

In the long run, a firm determines its level of output using the same fundamental profit-maximization rule as in the short run. The firm aims to produce at the level where its marginal revenue (MR) equals marginal cost (MC), since this is the point at which profit is maximized. In a perfectly competitive market, the firm is a price taker, meaning it cannot influence the market price and must accept it as given. Therefore, the firm's marginal revenue is equal to the market price.

$$MR = MC$$

Since:

$$MR = P$$

It follows that:

$$MC = P$$

This means that in the long run, the firm will adjust its output level until the marginal cost of producing an additional unit equals the market price. If the marginal cost is less than the price, the firm can increase profit by expanding production. If the marginal cost is greater than the price, the firm should reduce output to avoid losses. The equilibrium is reached when MC

equals price, ensuring that the firm is producing the most efficient level of output. Thus, even in the long run, the condition MC = P remains the key rule guiding a competitive firm's output decision.

Shutdown Decision in Long Run

The shutdown decision in the long run is relatively simple compared to the short run because all costs are variable and avoidable. Unlike the short run, there are no fixed or sunk costs that the firm is forced to bear. This means the firm has complete flexibility to adjust its inputs, scale of production, or even exit the market entirely. As a result, the firm must ensure that its total revenue is sufficient to cover all its costs in order to continue operating.

If a firm finds that it is incurring any economic loss, meaning its total revenue is less than its total cost, it will choose to shut down and exit the market. Continuing production in such a situation would not be rational, since the firm can avoid all costs by ceasing operations. Therefore, in the long run, only firms that are able to earn at least normal profit (where revenue equals total cost) will remain in the market. Firms that fail to do so will exit, ensuring that resources are allocated efficiently over time.

Long-Run Supply Curve

In the long run, the supply curve of a perfectly competitive firm is derived from its long-run marginal cost (LRMC) curve, but only the portion that lies above the long-run average cost (LRAC) curve. This is because, in the long run, all costs are variable and the firm must cover all its costs to remain in business. The firm will choose to produce only when the market price is at least equal to its average cost; otherwise, it will incur losses and exit the market.

A profit-maximizing firm determines its output where price equals marginal cost (P = MC). In the long run, this condition becomes P = LRMC, and the firm supplies output only if this price is also sufficient to cover the average cost. Therefore, the relevant portion of the LRMC curve is the segment that lies above the LRAC curve, as this ensures that the firm earns at least normal profit.

If the market price is below the LRAC, the firm cannot cover its total costs and will shut down in the long run. On the other hand, if the price is equal to the minimum point of the LRAC curve, the firm earns zero economic profit (normal profit) and continues operating. Thus, the

long-run supply curve reflects the firm's willingness to supply output at different price levels while ensuring long-term sustainability and efficiency.

Entry and Exit of Firms

In a perfectly competitive market, the number of firms in the industry is determined by the processes of entry and exit, which occur in response to profit or loss. Entry of firms takes place when existing firms in the market are earning supernormal (abnormal) profits. These high profits act as an incentive for new firms to enter the industry, as they expect to earn similar returns. As new firms enter, the total market supply increases, which leads to a fall in market price and a reduction in profits.

On the other hand, exit of firms occurs when firms in the market are incurring losses. Losses signal that resources are not being used efficiently, so some firms choose to leave the industry to avoid further losses. As firms exit, the total market supply decreases, which causes the market price to rise and reduces the losses of the remaining firms.

This continuous process of entry and exit ensures that, in the long run, firms earn only normal profit (zero economic profit). It also helps maintain equilibrium in the market by adjusting supply in response to changes in profitability.

Long-Run Equilibrium

In a perfectly competitive market, long-run equilibrium is achieved when firms reach a position where price equals marginal revenue, marginal cost, and long-run average cost:

$$P = MR = MC = LRAC$$

At this point, firms are producing at the most efficient level of output, where the cost of producing an additional unit (MC) is exactly equal to the revenue earned from it (MR), and this is also equal to the market price. Since price is equal to long-run average cost (LRAC), firms are able to cover all their costs, including both explicit and implicit (opportunity) costs.

One of the key implications of this equilibrium is that firms earn zero economic profit, also known as normal profit. This does not mean firms are making no money; rather, it means they are earning just enough to keep their resources in the current use, with no extra incentive for

new firms to enter or existing firms to leave the market. As a result, there is no tendency for entry or exit, and the market remains stable.

Another important implication is that the price equals the minimum point of the average cost curve, which indicates that firms are operating at the most efficient scale of production. This leads to an efficient allocation of resources, where goods are produced at the lowest possible cost and supplied at a price that reflects this efficiency. Thus, long-run equilibrium in perfect competition ensures both productive efficiency and overall economic efficiency in the market.

Adjustment Process

The adjustment process in a perfectly competitive market explains how the forces of entry and exit bring the market toward long-run equilibrium. When firms in the market are earning supernormal profits, it signals that the industry is highly profitable. This attracts new firms to enter the market, as they expect to earn similar profits. As more firms enter, the overall market supply increases, which leads to a fall in market price. This process continues until the extra profits are eliminated and firms earn only normal profit.

On the other hand, when firms are incurring losses, it indicates that the market is not profitable. As a result, some firms exit the market to avoid further losses. The exit of firms reduces the total market supply, which in turn causes the market price to rise. This rise in price helps the remaining firms recover their costs. The process of exit continues until losses are eliminated and firms again earn normal profit.

Thus, the adjustment process ensures that in the long run, the market stabilizes at a point where firms earn zero economic profit, with no incentive for further entry or exit.

Conclusion

In conclusion, the concept of perfect competition explains how firms operate in a market where they are price takers and have no control over market price. The firm’s demand curve is horizontal because it can sell any quantity at the prevailing price, which arises due to the presence of many buyers and sellers, identical products, and perfect information. The derivation of the demand curve through residual demand shows why individual firms face highly elastic demand.

The theory of profit maximization highlights that firms aim to produce at the level where marginal revenue equals marginal cost (MR = MC). In the short run, firms may earn profits, incur losses, or break even, but they continue production as long as they can cover variable costs. In the long run, firms can freely enter or exit the market, and this adjustment process ensures that all firms earn only normal profit.

Multiple Choice Questions

1. What defines a perfectly competitive market?

   A. All firms have the ability to influence market prices.

   B. There are many buyers and sellers, and products are identical.

   C. Firms operate with significant market power.

   D. Prices are set by individual firms based on demand.

   Answer: B. There are many buyers and sellers, and products are identical.

2. Why is a firm considered as a price taker in a perfectly competitive market?

   A. Because it has a unique product that no other firm offers.

   B. Because it can influence the market price through its output.

   C. Because the firm's output is insignificant relative to total market supply.

   D. Because it operates in a monopoly market structure.

   Answer: C. Because the firm's output is insignificant relative to total market supply

3. In the long run, what happens to firms in a perfectly competitive market that are earning supernormal profits?

   A. They will continue to earn supernormal profits indefinitely.

   B. Other firms will enter the market, driving down prices.

C. Firms will exit the market to avoid further losses.

D. They will increase their prices to maximize profits.

Answer: B. Other firms will enter the market, driving down prices.

4. What is the implication of the horizontal demand curve faced by a firm in perfect competition?

   A. The firm can raise its prices without losing customers.

   B. The firm can sell any quantity at the prevailing market price.

   C. Demand for the firm's product is inelastic.

   D. The firm has control over the market supply.

   Answer: B. The firm can sell any quantity at the prevailing market price

5. What is meant by long-run equilibrium in a perfectly competitive market?

   A. Firms earn supernormal profit due to high demand.

   B. Price equals long-run average cost, resulting in zero economic profit.

   C. Firms will always have the option to exit the market.

   D. The market will fluctuate continuously without stabilizing.

   Answer: B. Price equals long-run average cost, resulting in zero economic profit.